Hrithik Nandi

Personal Data

Date of Birth:	18 March 2000
Address (Work):	Institute for Advancing Intelligence (IAI) TCG CREST, Kolkata, India
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Research Positions Held

Aug. 2022 – Present	**Cryptology and Security Group**, IAI TCG CREST, Kolkata, India Position: Senior Research Fellow

Education

Aug. 2022 – Present	**Ph.D. in Computer Science** Institute for Advancing Intelligence (IAI), TCG CREST Ramakrishna Mission Vivekananda Educational and Research Institute Advisor(s): Dr. Avijit Dutta & Dr. Soumitra Samanta
Aug. 2020 – Jul. 2022	**M.Sc. in Mathematics** Ramakrishna Mission Vivekananda Educational and Research Institute
Jul. 2017 – Jun. 2020	**B.Sc. in Mathematics** Ramakrishna Mission Vivekananda Centenary College

Research Interests

- Provable Security
- Design and Analysis of Symmetric Key Cryptographic Schemes
- Post-quantum Security of Symmetric Key Cryptographic Schemes

Publications

1. Samir Kundu, Sougata Mandal, Hrithik Nandi. **Committing Security of Forkcipher-based AEAD**. *IACR Communications in Cryptology, Vol. 3, No. 3.*
DOI: 10.62056/aytxl83y6, ePrint: 2026/2479.

2. Nilanjan Datta, Avijit Dutta, Sougata Mandal, Hrithik Nandi, Amlan Sinha. **Post-Quantum Security of Keyed Sum of Permutations and Its Siblings**. *IACR ASIACRYPT 2026.*
To Appear, ePrint: 2026/426.

3. Nilanjan Datta, Hrithik Nandi, Soumit Pal, Yu Sasaki, Patrick Struck, Maximiliane Weishäupl. **Generic Committing Attacks: Zero-Padded Ascon is Less Secure than Expected**. *IACR CRYPTO 2026.*
DOI: 10.1007/978-3-032-35412-9_17, ePrint: 2026/1160.

4. Nilanjan Datta, Jean Paul Degabriele, Avijit Dutta, Vukašin Karadžić, Hrithik Nandi. **Rugged Pseudorandom Permutations with Beyond-Birthday-Bound Security**. *ACM ASIACCS 2026.*
DOI: 10.1145/3779208.3805985, ePrint: 2025/1089.

5. Ritam Bhaumik, Nilanjan Datta, Avijit Dutta, Ashwin Jha, Sougata Mandal, Bart Mennink, Hrithik Nandi, Yaobin Shen. **How to Build a Short-Input Random Oracle from Public Random Permutations**. *IACR EUROCRYPT 2026*.
DOI: 10.1007/978-3-032-25333-0_14, ePrint: 2026/336.

6. Nilanjan Datta, Avijit Dutta, Shibam Ghosh, Eik List, Hrithik Nandi. **HCTR+: An Optimally Secure TBC-based Accordion Mode**. *IACR Transactions on Symmetric Cryptology, 2025(3)*.
DOI: 10.46586/tosc.v2025.i3.183-229, ePrint: 2024/2053.

7. Nilanjan Datta, Avijit Dutta, Sougata Mandal, Hrithik Nandi. **Sequential Indifferentiability of STH and EDM**. *IACR Communications in Cryptology, Vol. 2, No. 2*.
DOI: 10.62056/a3n59qxqi, ePrint: 2025/1518.

8. Ritam Bhaumik, Wonseok Choi, Avijit Dutta, Cuauhtemoc Mancillas López, Hrithik Nandi, Yaobin Shen. **Efficient Variants of TNT with BBB Security**. *ProvSec 2024*.
DOI: 10.1007/978-981-96-0957-4_1, ePrint: 2024/1237.

Preprints

1. Sougata Mandal, Hrithik Nandi, Amlan Sinha. **Committing Security of BBB Secure MACs**.
ePrint: 2026/425.

2. Avik Chakraborti, Bishwajit Chakraborty, Nilanjan Datta, Avijit Dutta, Ashwin Jha, Sougata Mandal, Hrithik Nandi, Mridul Nandi, Abishanka Saha. **Naor-Reingold goes Beyond-the-Birthday-Bound**.
ePrint: 2025/1486.

Invited and Contributed Talks

Sep. 2026	**Generic Committing Attacks: Zero-Padded Ascon is Less Secure than Expected** GAPS 2026, TU Wien, Vienna, Austria
Mar. 2026	**HCTR+: An Optimally Secure TBC-based Accordion Mode** FSE 2026, Singapore
Mar. 2026	**Short-Input Random Oracle from Public Random Permutations** ASK 2026, NTU, Singapore
Dec. 2025	**Sequential Indifferentiability of STH and EDM** Crypto Winter School 2025, IIT Bhilai, India
Dec. 2024	**Optimally Secure TBC Based Accordion Mode** ASK 2024, TCG CREST, Kolkata, India
Sep. 2024	**Efficient Variants of TNT with BBB Security** ProvSec 2024, Griffith University, Gold Coast, Australia

Workshops and Conferences Attended

Sep. 21 – 24, 2026	Workshop on Generic Attacks and Proofs in Symmetric Cryptography (GAPS 2026), TU Wien, Vienna, Austria
Mar. 23 – 27, 2026	Fast Software Encryption (FSE 2026) Singapore, Singapore

Mar. 19 – 22, 2026	Asian-workshop on Symmetric Key Cryptography (ASK 2026) NTU, Singapore
Sep. 01 – 05, 2025	Workshop on Generic Attacks and Proofs in Symmetric Cryptography (GAPS 2025), NTU, Singapore
Dec. 14 – 17, 2024	Asian-workshop on Symmetric Key Cryptography (ASK 2024) Kolkata, India
Dec. 09 – 13, 2024	Asiacrypt 2024 Kolkata, India
Sep. 25 – 27, 2024	International Conference on Provable and Practical Security (ProvSec 2024), Gold Coast, Australia
Dec. 10 – 13, 2023	Indocrypt 2023 Goa, India
Dec. 11 – 14, 2022	Indocrypt 2022 Kolkata, India

Academic Services

External Reviewer	CRYPTO 2026 EUROCRYPT 2027
Teaching Assistant	Discrete Mathematics, IAI TCG CREST, Aug. 2024 – Dec. 2024 Cryptology, IAI TCG CREST, Aug. 2023 – Dec. 2023

References

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Declaration

I hereby declare that the information stated above is true to the best of my knowledge and belief.

Place: Kolkata, India Hrithik Nandi